Aislinn Coleman-Plante

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https://acoleman-plante.github.io

EDUCATION

University of Colorado Boulder
Bachelor of Arts, Major: Astronomy | Minor: French — May 2024
Honors: Magna cum laude — *GPA: 3.75*

WORK EXPERIENCE

Post-baccalaureate Research Fellow, National Radio Astronomy Observatory — **Sept 2024-Present**
Supervisor: John Tobin, PhD

Research Assistant, CU Boulder — **May 2023-May 2024**
Supervisor: Meredith MacGregor, PhD
Honors thesis presented Spring 2024, to be submitted for publication

Learning Assistant, Modern Physics: CU Boulder — **Spring 2022**
Topics: Special Relativity, Quantum Mechanics, Atomic Structure
Supervisor: Noah Finkelstein, PhD

PUBLICATIONS

Coleman-Plante A, **"Investigating the HD 139664 Debris Disk and Star with ALMA,"** Undergraduate honors thesis, 2024, University of Colorado Boulder.
https://scholar.colorado.edu/concern/undergraduate_honors_theses/5h73px85p

Coleman-Plante, A., Tobin, J., Sheehan, P., & Yesmin, N., **"Protostellar Multiplicity in the Taurus Molecular Cloud,"** Submitted for publication (2025). Available at: https://acoleman-plante.github.io/taurus_multiplicity_submitted.pdf

James Mason [..., Aislinn Coleman-Plante] et al., **"Coronal Heating as Determined by the Solar Flare Frequency Distribution Obtained by Aggregating Case Studies,"** Astrophysical Journal 2023.
https://doi.org/10.48550/arXiv.2305.05687
* Contributed through course-based research project; developed Python tools for baseline subtraction and flare energy analysis

PRESENTATIONS

AAS 247th Meeting, Phoenix AZ, January 2026, iPoster: "Characterizing protostellar accretion in Orion with JWST"

NRAO Student Research Symposium, Charlottesville VA, July 2025, Talk: "Multiplicity in the Early Phases of Star Formation: The Taurus Molecular Cloud"

AAS 245th Meeting, National Harbor MD, January 2025, iPoster: "Multiplicity in the Early Phases of Star Formation: The Taurus Molecular Cloud," *https://ui.adsabs.harvard.edu/abs/2025AAS...24524916C*

AAS 243rd Meeting, New Orleans LA, January 2024, iPoster: "Investigating the HD 139664 Debris Disk and Star with ALMA," *https://ui.adsabs.harvard.edu/abs/2025AAS...24524916C*

RESEARCH PROJECTS

Characterizing protostellar accretion in Orion with JWST *(in progress)* — **Sept 2025-Present**
Supervisor: John Tobin, PhD

- Extract atomic hydrogen recombination lines from NIRSpec and MIRI MRS IFU spectral cubes for 14 protostars.
- Use radiative transfer to model extinction and determine the intrinsic flux loss along each line of sight.
- Convert extinction-corrected line fluxes into accretion luminosities and accretion rates.

Protostellar Multiplicity Survey in the Taurus Molecular Cloud *(submitted for publication)* **Sept 2024-Present**

Supervisor: John Tobin, PhD

- Developed pipelines for calibrating and imaging high-volume datasets from ALMA and VLA
- Conducted in-depth analysis of protostar population statistics
- Served as PI on a Cycle 12 ALMA proposal to expand the survey to Flat Spectrum sources; proposal was ranked in the top quartile of distributed peer review

Weighing Young Stars in the Taurus Molecular Cloud *(in prep)* **Sept 2024-Present**

Supervisor: John Tobin, PhD

- Used ALMA carbon-isotopologue molecular line data to trace Keplerian gas kinematics and modeled emission with the *pdspy* radiative-transfer software to constrain protostellar masses.
- Fit a protostellar mass function for the sample and evaluated mass trends as a function of luminosity, evolutionary stage, and multiplicity to investigate early stellar mass assembly

Investigating the HD 139664 Debris Disk and Star with ALMA **Spring 2024**

Honors thesis, Supervisor: Meredith MacGregor, PhD

- Calibrated and reduced ALMA data using CASA
- Leveraged MCMC functions to model the assymetric architecture of this debris disk
- Iteratively measured stellar luminosity, leading to the discovery of the first millimeter flares from an F-type star

Evolution of Stellar Activity as a Function of Age **Spring 2024**

Presented in Observations & Instrumentation II, Professor: Kevin France, PhD

- Investigated five stars of varying ages with the Hubble Advanced Spectral Products program to analyze emission fluxes and track FUV evolution with stellar age
- Analyzed Hubble COS/STIS data and modeled emission lines to calculate luminosities
- Applied findings to an atmospheric mass-loss model for a 17 Myr hot Jupiter exoplanet

Multi-wavelength Observations of Flares on V1005 **Fall 2023**

Presented in Observations & Instrumentation, Professor: Zachary Berta-Thompson, PhD

- Collaborated in operating three telescopes at the Sommers-Bausch Observatory to identify stellar flares across different wavelengths and estimate temperatures using the Planck function
- Created pipeline to perform differential photometry and generate light curves in real-time

Modeling of Exoplanetary Transits in Multiple Wavelengths, using Titan Occultation Data **Spring 2023**

Presented in Research Methods in Astronomy, Professor: Paul Hayne, PhD

- Developed a versatile model to simulate multi-wavelength transits of a Titan-like exoplanet, enabling the analysis of methane-like behavior in light curves to assess habitability potential
- Programmed light curve variations for different orbital radii and conducted research on atmospheric effects of various stellar spectral types on the modeled exoplanet

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SKILLS

- Python
- Linux
- CASA
- LaTeX
- CARTA
- ds9
- Mathematica
- Conversational & written French
- SLURM

AWARDS & HONORS

Magna cum laude, CU Boulder **May 2024**

CU Boulder Chancellor's Achievement Scholarship **2019-2023**

Dean's List: Fall 2019, Spring 2020, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024

RELEVANT COURSEWORK

Formation & Dynamics of Planetary Systems | ASTR 3710

Observations & Instrumentation I & II | ASTR 3510 & 3520

Astrophysics I: Stellar & Interstellar | ASTR 3750

Planets, Moons, & Rings | ASTR 3750

Introductory Radio Astronomy
ASTR 5340 **audited at UVA**

Research Methods in Astronomy | ASTR 3400

Planets & Their Atmospheres | ASTR 3720

Astronomical Data Analysis | ASTR 3800

Becoming a Learning Assistant | EDUC 4610

Introduction to Astrochemistry
ASTR 5260 **audited at UVA**

OUTREACH & LEADERSHIP

Dark Skies, Bright Kids **Fall 2024-Present**

Volunteer-run science outreach group serving elementary schools across Virginia

- Helped create accessible astronomy lesson plans for blind and low-vision students & taught science classes for grades 3–12 at the Virginia School for the Deaf and the Blind
- Presented planetarium shows to elementary school students through an after-school library program

President, CU Boulder Rotaract Club **2023-2024**

Community service club in conjunction with Rotary Club

- Coordinated monthly volunteer events to distribute hot meals and clothing to Boulder's homeless population
- Organized a fundraiser and collaborated with a university in Istanbul to support the construction of a field hospital in Turkey following the devastating earthquakes
- Led efforts to plant native trees at homes impacted by the Colorado Marshall wildfires.

Secretary, CU Boulder Rotaract Club **2022-2023**